

A04 Reflection

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We decided to use backpropagation and studied it through an example based on a basketball academy. A robot called Bounce is trained to take free throws. Each section of the network is assigned a player: Bounce takes the shot, the hoop represents the correct answer, Ref Distance assesses the miss, Coach Byte works out what went wrong, and Asst. Coach Descend carries out the required corrections.

Basketball was suitable for an eleven-year-old since he had already experienced missing a shot and then making adjustments. The figures were most helpful; when we fired six shots, the loss fell from 1.0 to nearly zero, and the weight increased from 0.5 to about 1.0, and those figures made the situation clearer than words alone.

I was previously incorrect, and by drafting this, I am rectifying my error. Backpropagation determines who is to blame, and gradient descent carries out the correction. I gained a better understanding by trying to explain it to a child; when my explanation became unclear, it showed I still didn't fully understand it.

Olis put together the report and diagrams and explained all the details using only the five-step diagram and the loss chart.

For the next example, I should include a second neuron because, even though one weight is easy to grasp, real networks connect many neurons. Still, the basketball analogy worked, and I think an 11-year-old could understand backpropagation from that explanation; in fact, I understand it better now than I did at the beginning.

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Basketball made backpropagation easier for me to understand. Missing a shot, seeing what went wrong, and correcting it is a simple way to understand how a neural network learns.

The six-shot example helped me the most. The loss kept getting smaller while the weight moved closer to the correct value. Seeing the numbers change made the learning process clearer.

I also understood better that backpropagation and gradient descent are different. Backpropagation finds where the error came from, while gradient descent uses that information to adjust.

Explaining the topic to an 11-year-old also helped me think about the concept more simply. If I could not explain a part clearly, it meant I still needed to understand it better.

My contribution was creating the project's instruction document. Doing that also helped me organize the idea and understand how to explain it simply.

Overall, I understand backpropagation better now because the basketball example connected a difficult deep learning concept to something familiar and easy to picture.

Olis Bahari

Reflection: Coach Byte's Basketball Academy Teaching

Backpropagation

The aid referred to here, Coach Byte's Basketball Academy, appears in both the slide deck and the report; it features a training robot called 'Bounce,' which is used to explain backpropagation to a young or non-technical audience. The slide deck and the report are the same teaching, since the revisions made them part of the same project.

What Works Well

The basketball analogy does not merely serve as a decorative element since it is, in fact, carrying out real conceptual work; it enables the reader to get an intuitive understanding of the concepts before any mathematics is brought in by comparing a forward pass to a shot, a miss to how far it fell short of the hoop, and backpropagation to 'working out who is to blame' for the miss. The way of drawing the line between backpropagation ('the messenger: it calculates blame') and gradient descent ('acts on that message') is truly a clarifying method of distinguishing two ideas which even professionals often confuse.

From an educational point of view, the way the material is presented is logical since it starts with a specific example, then proceeds to provide a numerical example concerning six "shots" with the loss falling in each case, afterward applies the same mode of reasoning in a 12-line Python loop, and finally examines some real-world applications. The approach provides clear progression from an intuitive grasp of the concept through to coding and then to practical use. It's a valid point that the curve decreases by the same fraction at each stage since this reflects the actual dynamics of training and doesn't need a large model to show it.

The slide deck and the report are well coordinated because the section numbers, the Bounce example, and the code block match in both, letting readers move from the slides to the full report without feeling like they're getting two different explanations.

What Could Be Tightened

Many of the sentences in the report do more grammatical work than is needed. As written, 'when a neuron is active, it also adds bias confuses people about what bias is, since bias is added no matter whether the neuron is active and is applied before the activation function. Although the statement 'each connection has a weight that determines the strength of the signal' is acceptable, the report could be made clearer by considering 'signal strength' and 'the neuron's own bias as two separate and constantly present components rather than implying that the former depends on activity.

Overall Takeaway

It serves as a clear instance of employing a narrative approach which is easy to understand in order to explain a technically correct idea—namely, the calculation of gradients by means of the chain rule—without simplifying it so much as to lose accuracy; in fact, the way in which the "blame assignment" aspect is handled at the end demonstrates a correct and accurate grasp of what backpropagation achieves at each layer.